Tathagat Agrawal

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ML researcher and scientist with experience building production ranking and decision systems (learning-to-rank recommendation engines, LLM-based evaluation pipelines) and a growing focus on quantitative and algorithmic trading — including an event-driven backtesting engine with realistic execution modeling and full risk/performance analytics. Published 3 peer-reviewed papers. Seeking roles at the intersection of quantitative research and applied ML.

Work Experience

Applied Scientist (AI/ML) **October 2025 – Present**
Neural Nurture *Indore, MP, India*

- Developed a learning-to-rank **recommendation system (LambdaMART/XGBoost)** for the travel industry, engineering custom relevance features and **improving NDCG@3 by about 40%** over the prior baseline across multi-query test sets.
- Integrated **explainability techniques** (e.g. SHAP, attention weights) into recommendation models to surface human-interpretable explanations for end users.
- **Fine-tuned LLMs** using SFT and LoRA to generate natural language explanations for recommendations; diagnosed and resolved a training data bias causing hallucinations, **improving factual accuracy to 98.4%**.
- Built and trained a **multi-task NER + intent classification model** (RoBERTa backbone) on a custom dataset of 100,000 annotated samples, covering 44 entity classes, achieving an F1-score of over 94%.
- Designed **agentic tool-calling architecture** for a conversational travel application – including prompt engineering and external API integration (handling ToS compliance and cost optimization) – to route structured NER output to the right tools in the pipeline.
- Designed an **end-to-end evaluation framework** for the product, leading to early bug discovery and continuous monitoring of performance metrics.
- Translated recent **ML research** into production-ready prototypes by systematically reviewing papers, and **designing controlled experiments** across LLMs and conventional ML models to validate applicability to product use cases.
- Presented experimental results and architectural proposals to senior leadership, influencing product roadmap decisions based on empirical data; conducted technical interviews for ML internship candidates.

Business Analyst Intern **May 2024**
Finlatics – NITI Aayog, Govt. of India *Remote*

- Conducted case studies on data-driven decision-making.
- Automated data cleaning pipelines with Python scripts, significantly reducing processing time.
- Facilitated data-driven reporting to senior leadership.

Publications

[1] Agrawal T, Kumar M (2025) Grokking in Neural Networks: A Review. SN Computer Science 6:627. DOI link

[2] Agrawal T, Kumar M (2025) Analysis of Grokking-Induced Rigidity in Neural Networks using Transfer Learning. In: Proceedings of Data Analytics and Management, ICDAM.

[3] Agrawal T, Chauhan A, Singh A (2024) Advancing Linguistic Frontiers with mBERT Fine-Tuning for Hindi and English NER Using HiNER and WikiNEuRal Datasets. In: ICCCNT 2024. Paper link. F1 of 0.899 (HiNER), 0.946 (WikiNEuRal).

Technical Skills

Languages: Python, C/C++, Java, JavaScript, R
ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, HuggingFace Transformers, OpenCV, MLflow, XGBoost
LLMs & NLP: LangChain, LLaMA-Factory, OpenAI API, spaCy, NLTK
Deployment & APIs: FastAPI, Flask, REST APIs, Azure ML
Data & Visualisation: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel
Developer Tools: Git, GitHub, Claude Code, Cursor, LaTeX

Education

Integrated M. Tech. in AI **2020 – 2025**
VIT Bhopal University, Bhopal *CGPA: 9.1/10.0*

- **Relevant courses:** Applied ML, Computer Vision, NLP, Deep Learning, Hardware Acceleration for ML, Data Visualization, Big Data and Data Warehousing, Probability and Statistics, etc.
- **Master's thesis:** *Investigating the Effects of Grokking Phenomenon in Transfer Learning Settings.*
  - Designed and performed transfer learning experiments using MLP and Transformer models to study the effects of grokking on model flexibility using loss landscape visualization and Hessian eigenvalue analysis.
  - See publications [1] and [2].

Projects

Strategy Backtester **August 2026**

Python, pandas, NumPy, CCXT, Typer, Rich, pytest, Matplotlib GitHub

- Built an event-driven backtesting engine from scratch, with realistic slippage and transaction-cost modeling and structural lookahead-bias prevention enforced by a dedicated regression suite.
- Implemented order-execution logic including limit/stop fills and bracket (OCO) take-profit/stop-loss orders, plus a same-bar-close comparison mode to quantify realistic vs. naive fill timing.
- Built a CCXT-based market-data collector and a full performance-analytics suite (Sharpe, Sortino, CAGR, max drawdown, win rate, cost transparency), validated end-to-end against 9 months of real BTC/USDT hourly data.
- Shipped two trading strategies (SMA crossover, RSI mean-reversion) behind a Typer/Rich CLI supporting CSV/JSON export and equity-curve/drawdown chart plotting.

AI Clinic Scheduler **August 2026**

LangGraph, LangChain, FastAPI, Pydantic, ChromaDB, Sentence-Transformers, OpenRouter, HTML/JS GitHub

- Built an LLM-powered appointment scheduling agent using a LangGraph ReAct agent to hold multi-turn conversations, interpret patient symptoms, and recommend a doctor via retrieval-augmented generation over a knowledge base of specialties and bios.
- Implemented a RAG-based doctor-matching pipeline with Sentence-Transformers embeddings persisted in a Chroma vector store, exposed to the agent as a callable retriever tool.
- Designed explicit Pydantic schemas for every tool call so malformed arguments from less-reliable models surface as correctable errors instead of crashing the agent.
- Built a FastAPI backend serving patient chat, doctor/slot lookups, and admin endpoints, alongside a no-build-step HTML/JS frontend for patient booking and a receptionist admin view.

Multilingual NER with mBERT **December 2023**

PyTorch, HuggingFace, Fine-tuning, NLP Paper link

- Fine-tuned the bert-base-multilingual-cased model for Named Entity Recognition for Hindi (HiNER) and English (WikiNEuRal).
- Pre-processed datasets via subword tokenization and aligned entity annotations to token IDs. Conducted hyperparameter-tuning using grid search to obtain optimal performance.
- Achieved F1 = 0.899 (HiNER) and F1 = 0.946 (WikiNEuRal), outperforming Indic-BERT and XLM-R baselines (see publication [3]).

Star Classification using Supervised Learning Models **August 2025**

Scikit-learn, MLflow, Pandas, Matplotlib, NumPy, Flask, HTML GitHub

- Trained and evaluated MLP, SVC, and Random Forest classifiers for six class stellar classification with rigorous hyperparameter tuning.
- Used MLflow for experiment tracking, hyperparameter logging, and automatic selection of best-performing model.
- Implemented a Flask web application for interactive predictions, including H-R diagram visualization.

Semantic Segmentation using U-Net **July 2025**

PyTorch, U-Net, CNN, Segmentation GitHub

- Implemented the U-Net architecture for binary semantic segmentation on the Oxford-IIIT Pet dataset.
- Designed and optimized data pre-processing pipelines (augmentation, normalization, resizing) and trained the network with binary cross-entropy loss and Adam optimizer.
- Achieved high segmentation accuracy of 91.89%.

Multiple Disease Prediction in Crops using Deep Learning **December 2022**

CNN, PyTorch, OpenCV, Keras GitHub

- Implemented a stacked CNN to classify crop leaves into one of 38 disease categories.
- Pre-processed images via resizing, normalization, and data augmentation and mitigated class imbalance to prevent overfitting.
- Trained a multi-block CNN architecture followed by fully connected layers with dropout and L2 regularization using Adam on cross-entropy loss, achieving 98.28% test accuracy and more than 98% precision, recall, and F1-score.

Certifications

Neural Networks and Deep Learning **May 2021**

DeepLearning.AI, Coursera Certificate link

Mathematics for Machine Learning **May 2020**

Imperial College London, Coursera Certificate link